Rapid Approximate Spatial Prediction Method

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Background

In spatial statistics, the observational model can be written as:

$$y_i = \mu\left(\mathbf{s}_i^O\right) + g\left(\mathbf{s}_i^O\right) + \epsilon_i, \text{ for } i = 1, \ldots, n$$

Here, y_i is the i-th observation at the i-th location $\mathbf{s}_i^O$ and ϵ_i is its associated irreducible noise. Suppose we want $\hat{y}$ at unknown gridded locations $\mathbf{s}_k^G$. If the fixed effect $\mu(\cdot)$ can be modeled using regression, then a mean zero Gaussian spatial process is estimated by *universal kriging* as:

$$\hat{g}_{exact}\left(\mathbf{s}_k^G\right) = \sum_{i=1}^{n} k\left(\mathbf{s}_k^G, \mathbf{s}_i^O\right)\mathbf{c}_i \qquad (1)$$

where

$$\mathbf{c} = (\mathsf{K} + \sigma^2 I)^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \qquad (2)$$

where $k\left(\mathbf{s}_k^G, \mathbf{s}_i^O\right)$ is the covariance between $\mathbf{s}_k^G$ and $\mathbf{s}_i^O$, K is the cross-covariance between each observation locations, σ^2 the nugget variance, $\mathbf{X}$ matrix of covariates and $\hat{\boldsymbol{\beta}}$ the GLS estimate.

Evaluating (1) can require excessive time, making it difficult to

- perform interactive analysis and modeling of spatial data,
- quickly evaluate fitted surfaces for conditional simulation [1], and more.

The rapid approximate kriging prediction method

- requires stationary covariances,
- leverages nearest neighbor grid points to estimate off-grid covariances,
- uses sparse linear algebra and convolution via FFT.

We show case its accuracy and speed through an example using North American annual summer rainfall data along the 100th meridian.

The Method

Estimating $k\left(\mathbf{s}_k^G, \mathbf{s}_i^O\right)$ via interpolation from nearest neighbors.

For a prediction grid of size M and $k, j \in (1, \ldots, M)$:

$$k\left(\mathbf{s}_k^G, \mathbf{s}_i^O\right) \approx \sum_{j \in \mathsf{M}(i)} k\left(\mathbf{s}_k^G, \mathbf{s}_j^G\right) A_{i,j} \qquad (3)$$

where A is a sparse matrix and $\mathsf{M}(i)$ contain indexes for nearest neighbors to $\mathbf{s}_i^O$. The non-zero elements of A are $\Sigma_*(\Sigma_{NN})^{-1}$ where, Σ_* is the cross-covariance between each observation and its nearest neighbors, and Σ_{NN} is the covariance between neighbors in the same neighborhood.

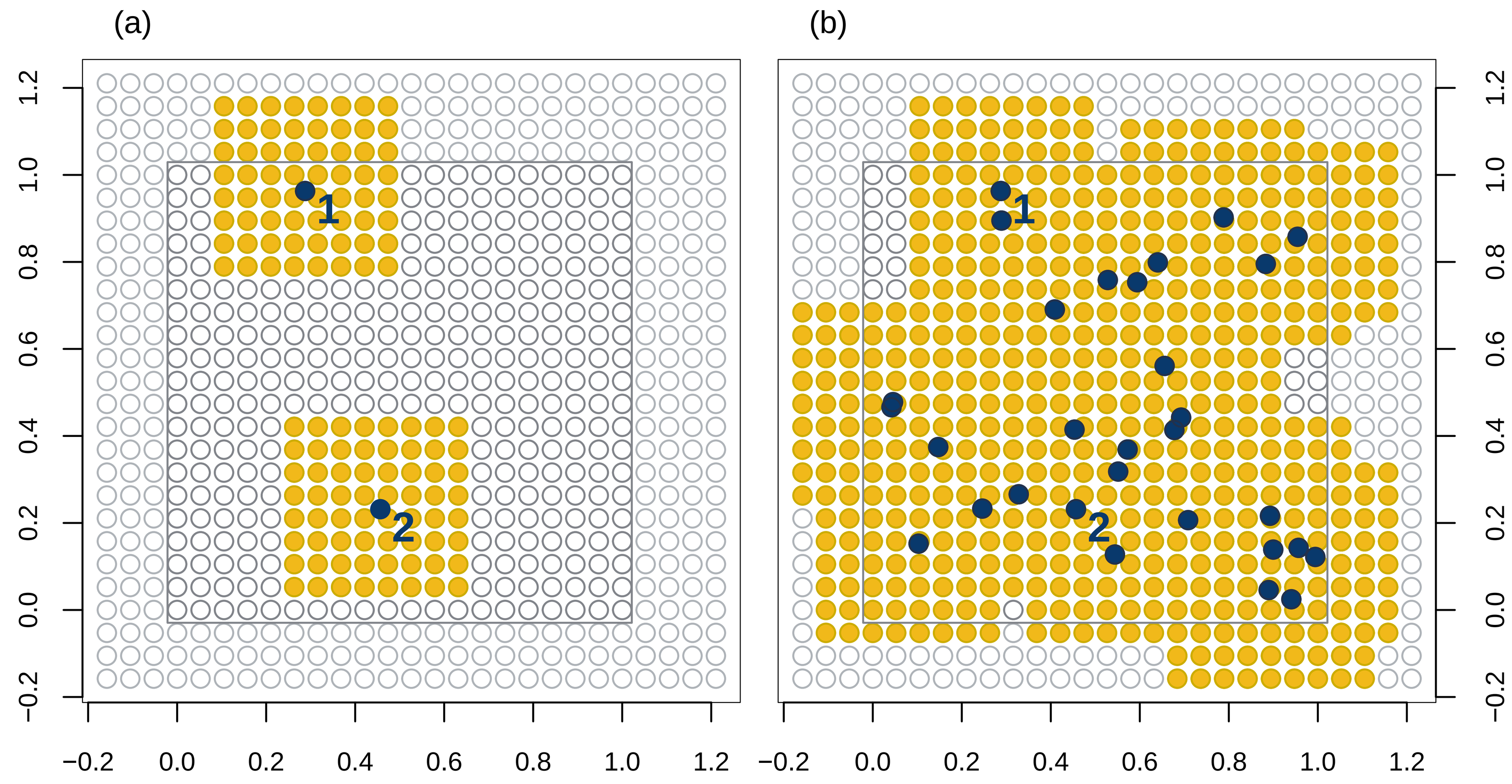

Figure 1. Example with 30 observations. The prediction grid of interest is in white with silver outlines, enclosed by padding in white with light gray outlines. Observations are labeled in dark blue and each observation's nearest neighbors $\mathsf{M}(i)$ in gold.

FFT, FFT, FFT

$$\hat{g}_{approx}(\mathbf{s}_k^G) \approx \sum_{i=1}^{n}\left[\sum_{j \in \mathsf{M}(i)} k\left(\mathbf{s}_k^G, \mathbf{s}_j^G\right) A_{i,j}\right]\mathbf{c}_i = \sum_{j=1}^{M} k\left(\mathbf{s}_k^G, \mathbf{s}_j^G\right)\mathbf{c}_j^* = \sum_{j=1}^{M} \Phi\left(\mathbf{s}_k^G - \mathbf{s}_j^G\right)\mathbf{c}_j^* \qquad (4)$$

where the new coefficients are $\mathbf{c}_j^* = \sum_{i=1}^{n} A_{i,j}\mathbf{c}_i$ when $j \in \mathbf{M} = \{\mathsf{M}(i) \mid i = 1, \ldots, n\}$ and $\mathbf{c}_j^* = 0$ when $j \notin \mathbf{M}$.

So, for $\{\Sigma_{full}\}_{i,j} = k(\mathbf{s}_k^G, \mathbf{s}_j^G) = \Phi\left(\mathbf{s}_k^G - \mathbf{s}_j^G\right)$ the full-grid approximate spatial effect is a convolution over a 2D array:

$$\hat{g}_{approx}\left(\mathcal{S}^G\right) = \Sigma_{full} * \mathbf{c}^*$$

which we speed up using the FFT.

Example: Prediction

The annual Summer rainfall dataset has $n = 1368$ observations along the 100th meridian of North America, with most observations inside the United States.

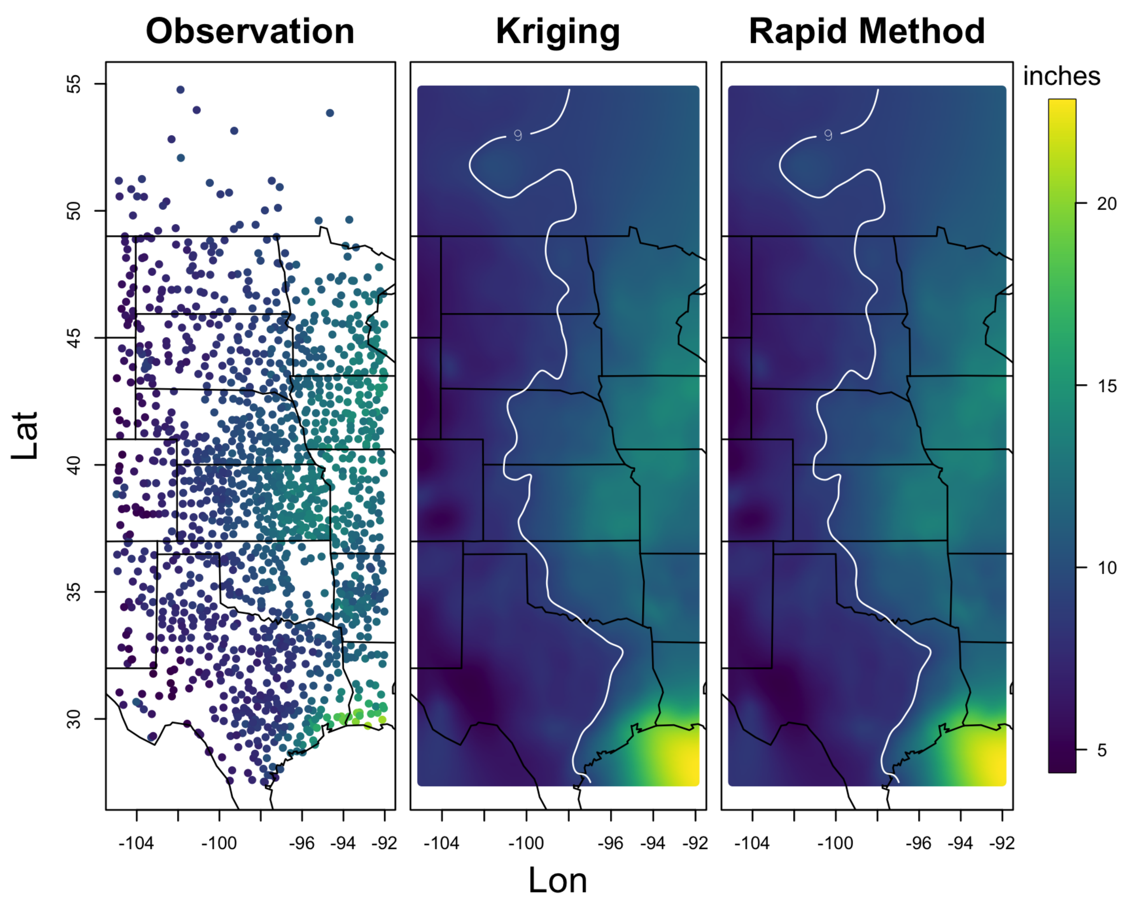

Figure 2. Kriging and rapid approximate kriging prediction results, both with contour line at 9 inches of rain. The prediction grid is 256×512.

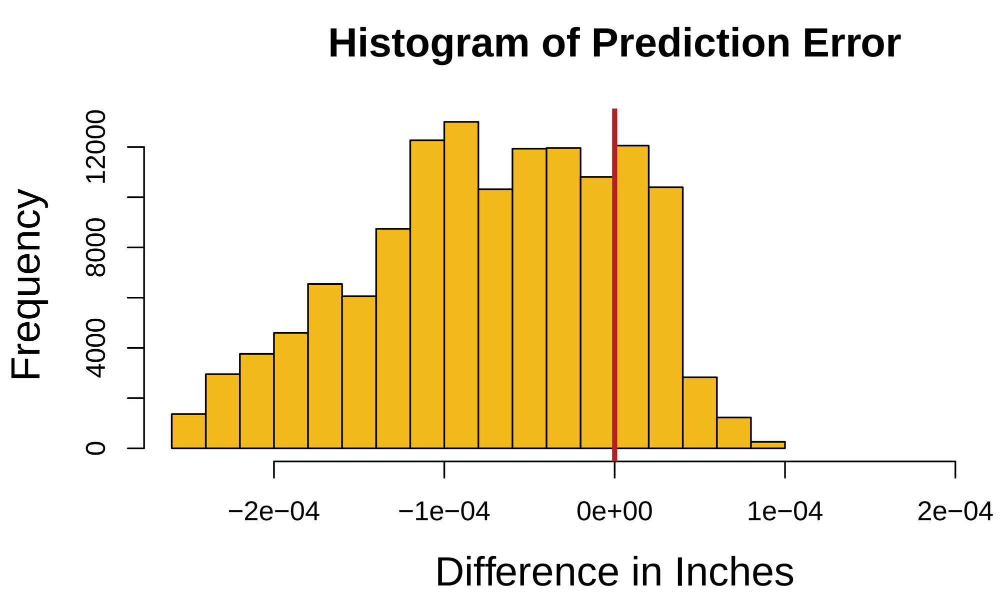

Figure 3. The difference between kriging and the rapid method from plot above. The errors are within $\pm 3 \times 10^{-4}$, which has been consistent with other examples and test cases.

Example: Timing & Conditional Simulation

All timing experiments are done on a laptop with an Apple M2 processor using Basic Linear Algebra Subprograms (BLAS) for Macs, and same rainfall data.

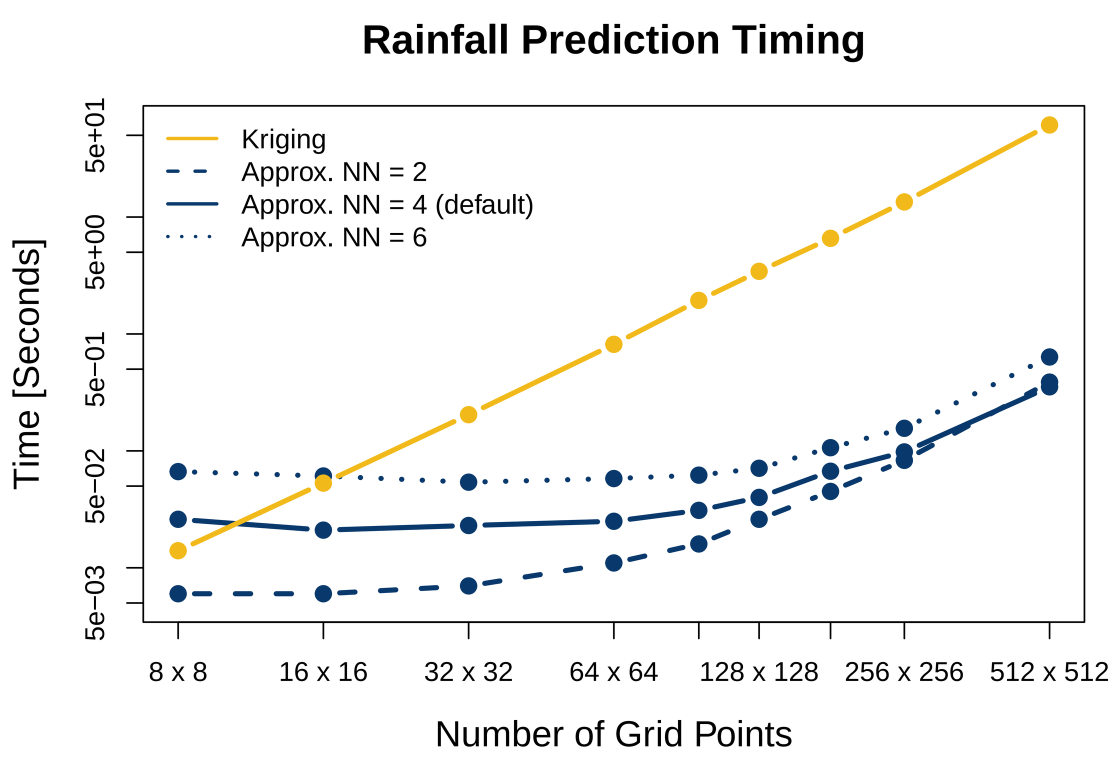

Figure 4. Each data point is taken from the median of 20 timing runs.

When implemented in the Conditional Simulation (CS) algorithm with Circulant Embedding (CE), computational time is drastically improved.

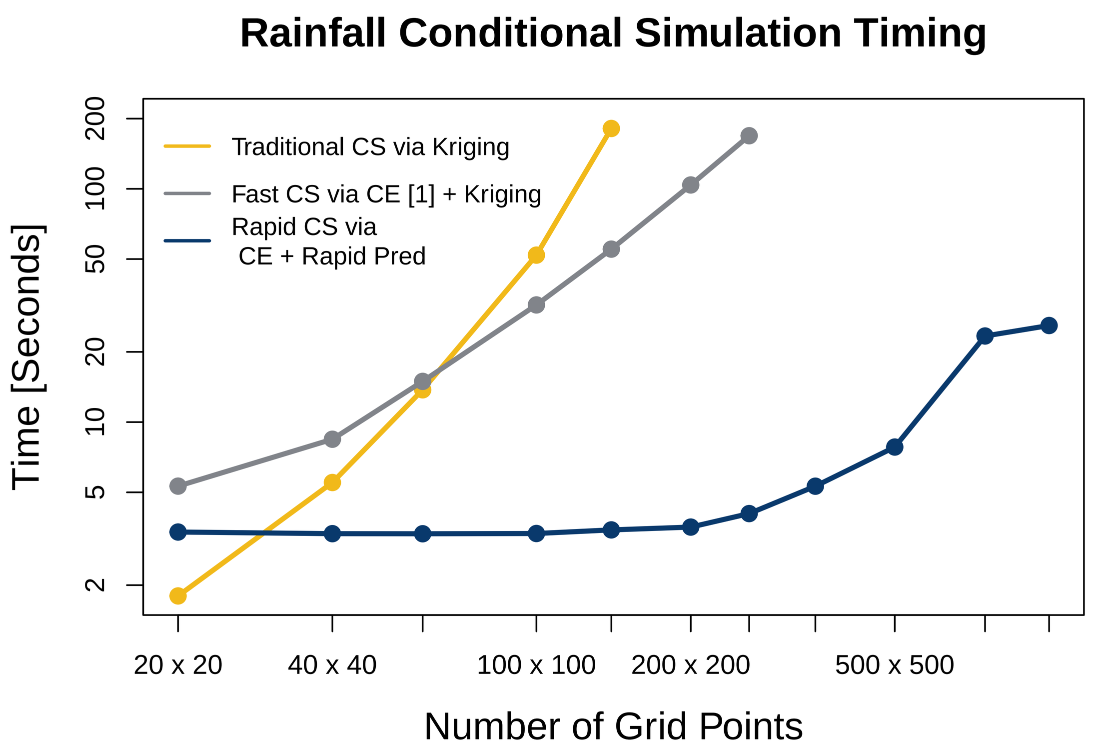

Figure 5. Timing for 10 ensemble simulations. The exact calculation for SE stops working around 750×750 grid points.

Summary

The rapid approximate spatial prediction method speeds up (1) from an order $O(nM)$ operation to an $O(M \log M)$ one with extreme accuracy.

- This method is superior in speed at any grid size.
- When implemented in conditional simulation algorithms, we see we see a hundred-fold speed up.
- When nearest neighbor size increases to be the full grid, this method is exactly the same as kriging.
- Nearest neighbor size of 4 is recommended.
- When implemented in CS, the approximation of the true standard error (SE) is accurate.

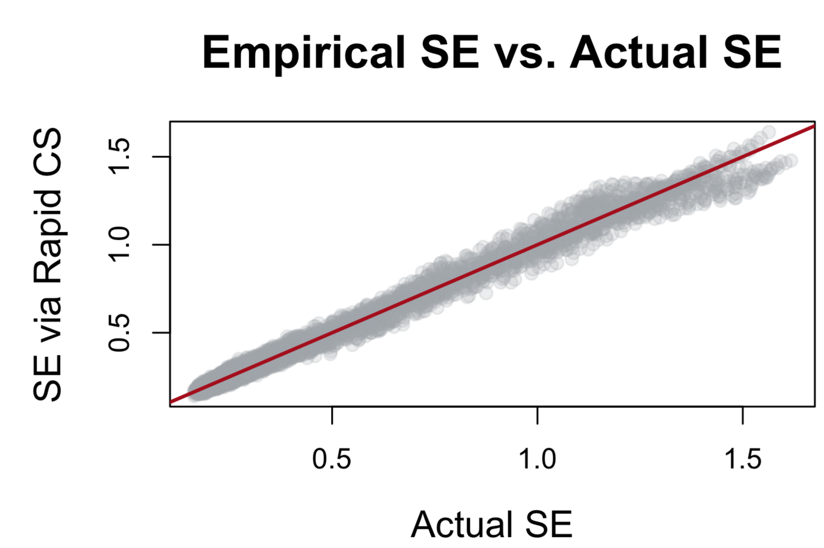

Figure 6. Empirical SE from 100 ensembles for rainfall example.

Future Work

- Tight analytical error bound via interpolation theory [3].
- Embed in highly composite grid sizes, more than 2D grid, and polished version in the `fields` R package [2].
- Regularize CE to always give a covariance matrix.
- Non-stationary models with weighted mixture of stationary covariances.

References

[1] M. D. Bailey, S. Bandyopadhyay, and D. Nychka. Adapting conditional simulation using circulant embedding for irregularly spaced spatial data. *Stat*, 11, 12 2022.

[2] Douglas Nychka, Reinhard Furrer, John Paige, and Stephan Sain. fields: Tools for spatial data, 2021. R package version 16.2.

[3] G. E. Fasshauer. *Meshfree approximation methods with MATLAB*. World Scientific, 2007.